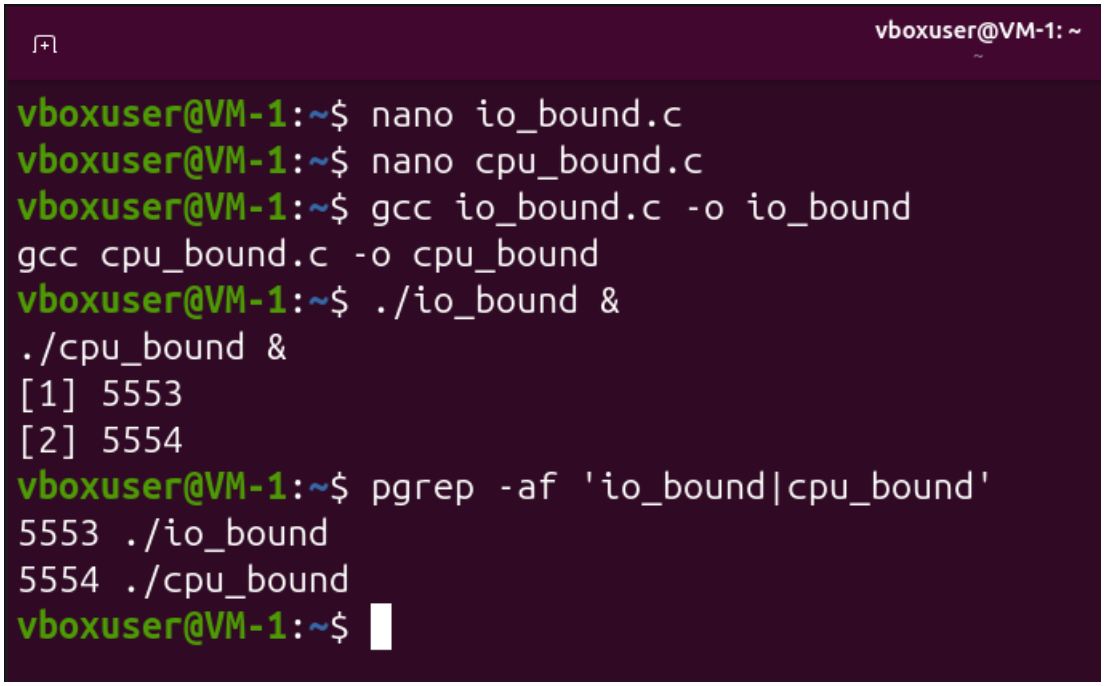
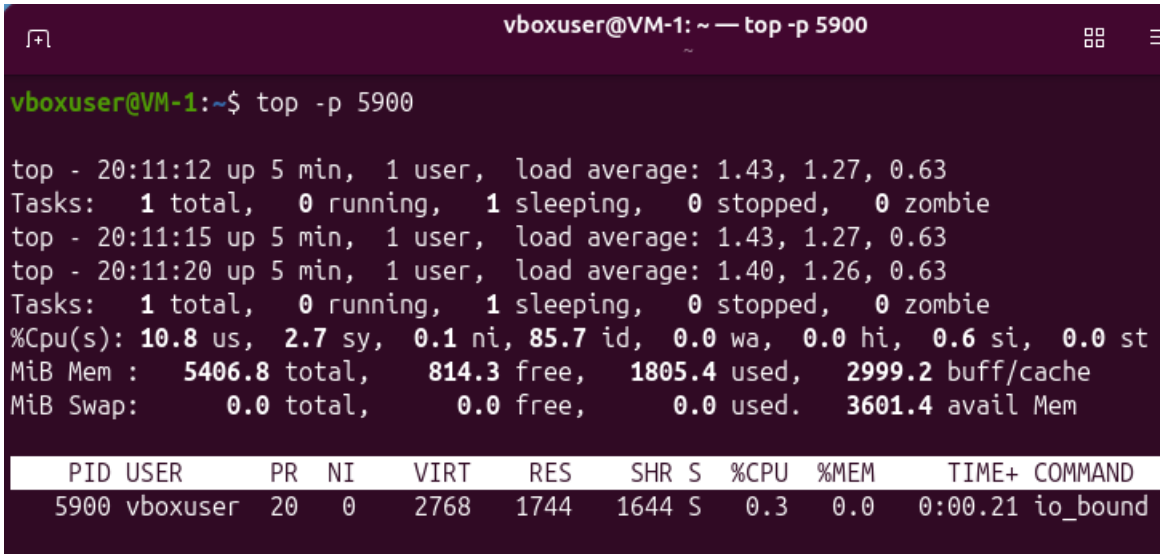

vboxuser@VM-1: ~
vboxuser@VM-1:~$ nano io_bound.c
vboxuser@VM-1:~$ nano cpu_bound.c
vboxuser@VM-1:~$ gcc io_bound.c -o io_bound
gcc cpu_bound.c -o cpu_bound
vboxuser@VM-1:~$ ./io_bound &
./cpu_bound &
[1] 5553
[2] 5554
vboxuser@VM-1:~$ pgrep -af 'io_bound|cpu_bound'
5553 ./io_bound
5554 ./cpu_bound
vboxuser@VM-1:~$


vboxuser@VM-1: ~ — top -p 5900
vboxuser@VM-1:~$ top -p 5900
top - 20:11:12 up 5 min, 1 user, load average: 1.43, 1.27, 0.63
Tasks: 1 total, 0 running, 1 sleeping, 0 stopped, 0 zombie
top - 20:11:15 up 5 min, 1 user, load average: 1.43, 1.27, 0.63
top - 20:11:20 up 5 min, 1 user, load average: 1.40, 1.26, 0.63
Tasks: 1 total, 0 running, 1 sleeping, 0 stopped, 0 zombie
%Cpu(s): 10.8 us, 2.7 sy, 0.1 ni, 85.7 id, 0.0 wa, 0.0 hi, 0.6 si, 0.0 st
MiB Mem : 5406.8 total, 814.3 free, 1805.4 used, 2999.2 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3601.4 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
5900 vboxuser 20 0 2768 1744 1644 S 0.3 0.0 0:00.21 io_bound

```
  GNU nano 8.7.1
#include <stdio.h>

int main()
{
    long long count = 0;

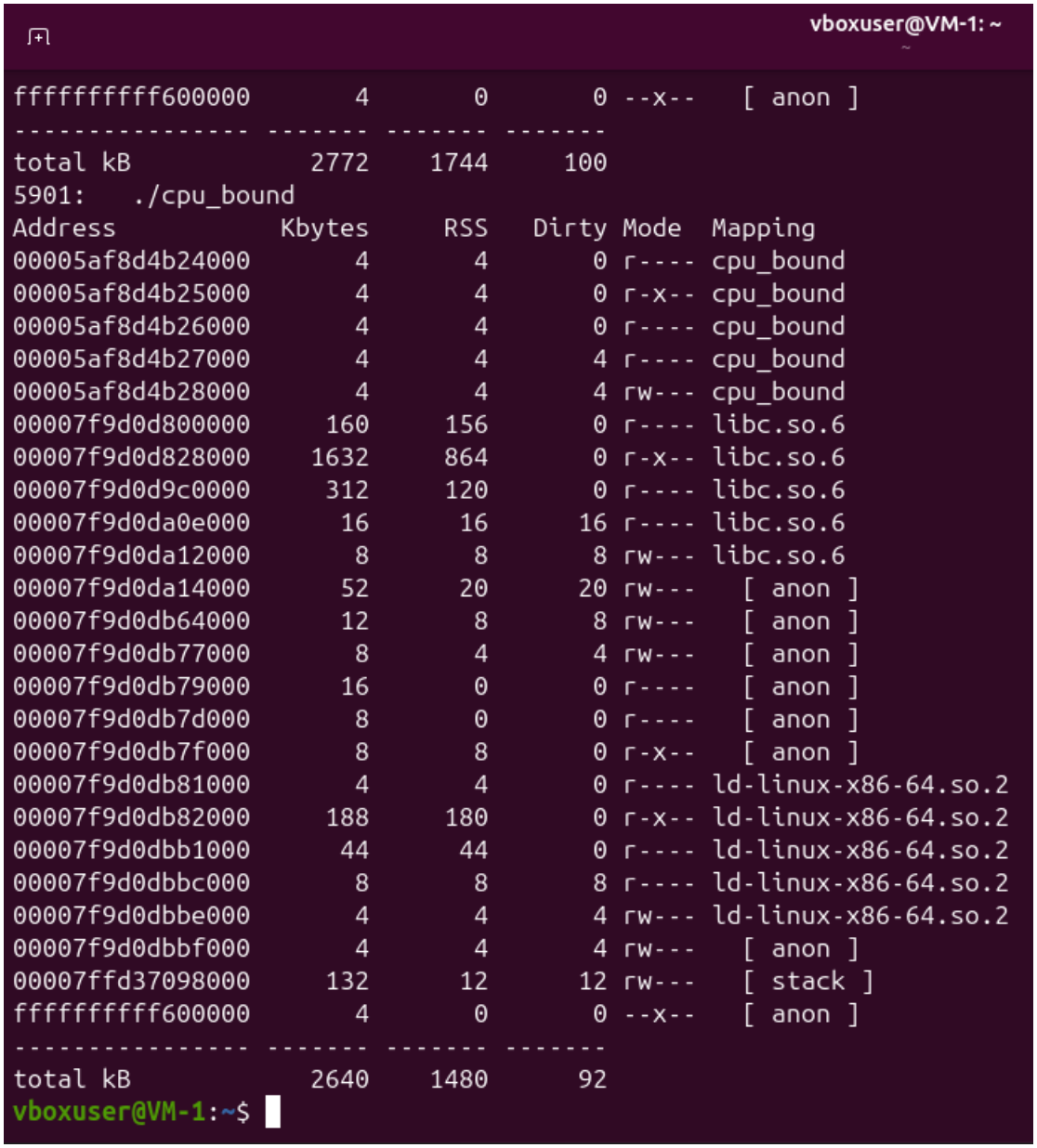
    while (1)
    {
        count++;
    }

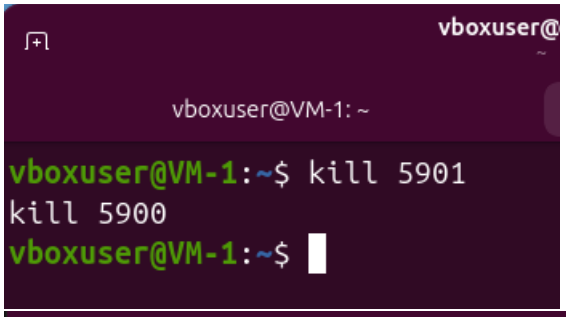
    return 0;
}
```

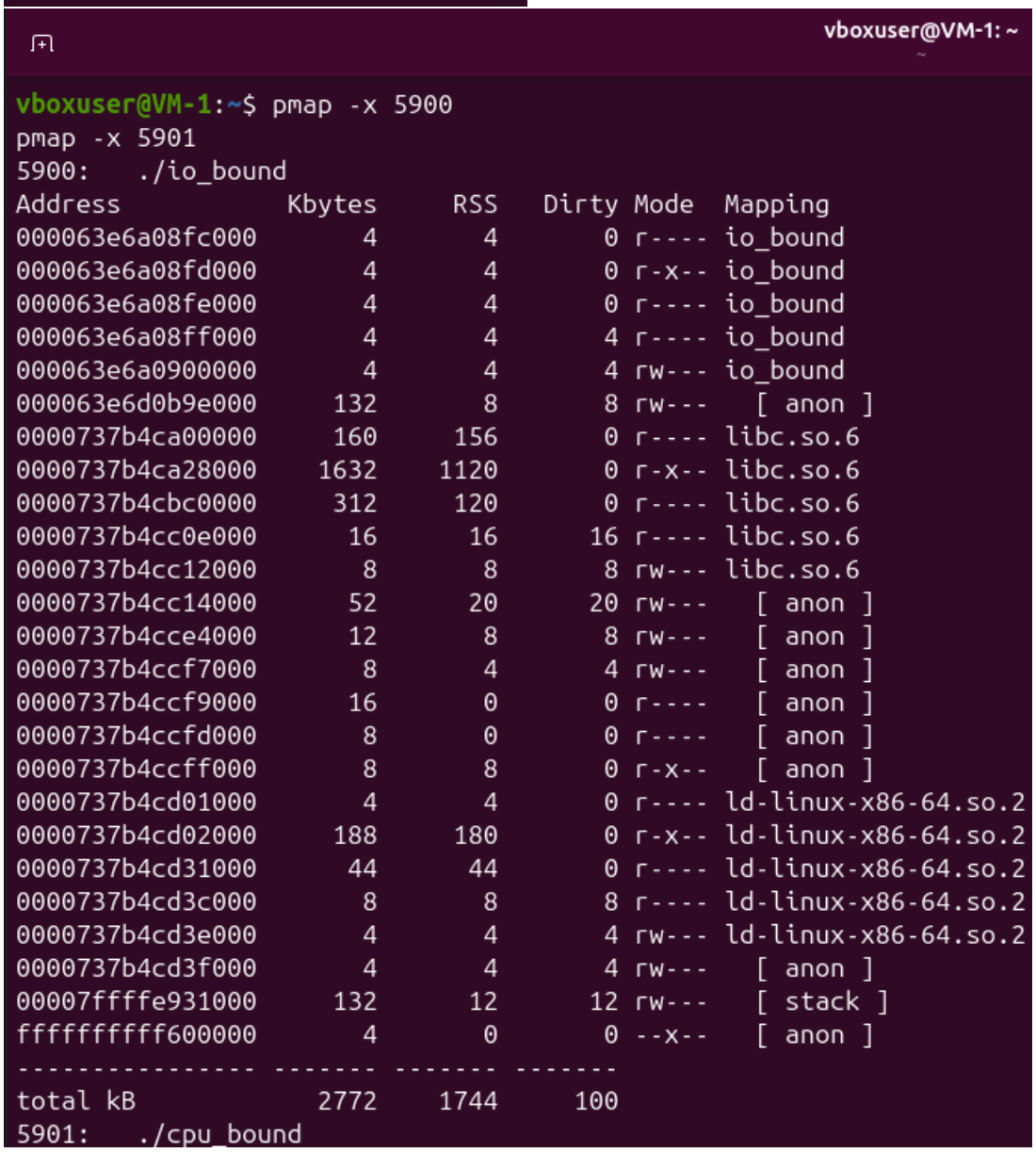
```
vboxuser@V
  GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>

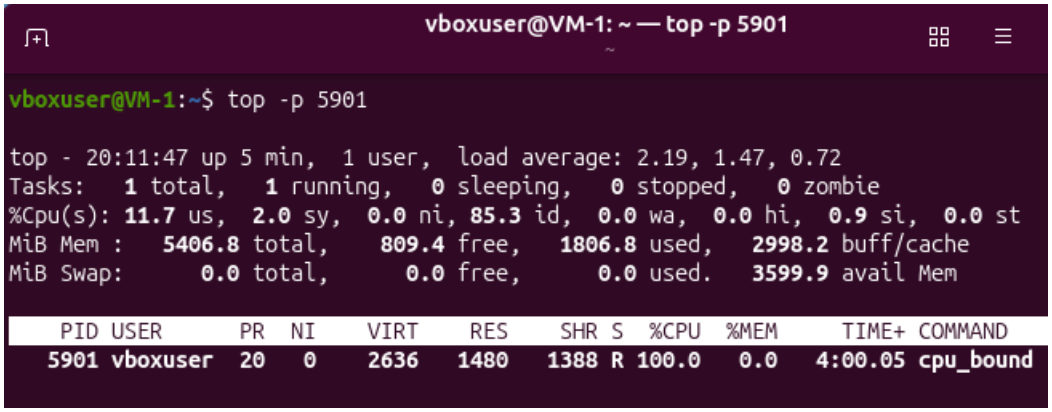
int main()
{
    FILE *fp;

    while (1)
    {
        fp = fopen("dummy.txt", "w");
        if (fp != NULL)
        {
            fprintf(fp, "Writing data\n");
            fclose(fp);
        }
        sleep(1);
    }
    return 0;
}
```


vboxuser@VM-1: ~
ffffffffff600000 4 0 0 --x-- [anon]
---------- ------- ------- -------
total kB 2772 1744 100
5901: ./cpu_bound
Address Kbytes RSS Dirty Mode Mapping
00005af8d4b24000 4 4 0 r---- cpu_bound
00005af8d4b25000 4 4 0 r-x-- cpu_bound
00005af8d4b26000 4 4 0 r---- cpu_bound
00005af8d4b27000 4 4 4 r---- cpu_bound
00005af8d4b28000 4 4 4 rw--- cpu_bound
00007f9d0d800000 160 156 0 r---- libc.so.6
00007f9d0d828000 1632 864 0 r-x-- libc.so.6
00007f9d0d9c0000 312 120 0 r---- libc.so.6
00007f9d0da0e000 16 16 16 r---- libc.so.6
00007f9d0da12000 8 8 8 rw--- libc.so.6
00007f9d0da14000 52 20 20 rw--- [anon]
00007f9d0db64000 12 8 8 rw--- [anon]
00007f9d0db77000 8 4 4 rw--- [anon]
00007f9d0db79000 16 0 0 r---- [anon]
00007f9d0db7d000 8 0 0 r---- [anon]
00007f9d0db7f000 8 8 0 r-x-- [anon]
00007f9d0db81000 4 4 0 r---- ld-linux-x86-64.so.2
00007f9d0db82000 188 180 0 r-x-- ld-linux-x86-64.so.2
00007f9d0dbb1000 44 44 0 r---- ld-linux-x86-64.so.2
00007f9d0dbbc000 8 8 8 r---- ld-linux-x86-64.so.2
00007f9d0dbbe000 4 4 4 rw--- ld-linux-x86-64.so.2
00007f9d0dbbf000 4 4 4 rw--- [anon]
00007ffd37098000 132 12 12 rw--- [stack]
ffffffffff600000 4 0 0 --x-- [anon]
---------- ------- ------- -------
total kB 2640 1480 92
vboxuser@VM-1:~$


vboxuser@
vboxuser@VM-1: ~
vboxuser@VM-1:~$ kill 5901
kill 5900
vboxuser@VM-1:~$


vboxuser@VM-1: ~
vboxuser@VM-1:~$ pmap -x 5900
pmap -x 5901
5900: ./io_bound
Address Kbytes RSS Dirty Mode Mapping
000063e6a08fc000 4 4 0 r---- io_bound
000063e6a08fd000 4 4 0 r-x-- io_bound
000063e6a08fe000 4 4 0 r---- io_bound
000063e6a08ff000 4 4 4 r---- io_bound
000063e6a0900000 4 4 4 rw--- io_bound
000063e6d0b9e000 132 8 8 rw--- [anon]
0000737b4ca00000 160 156 0 r---- libc.so.6
0000737b4ca28000 1632 1120 0 r-x-- libc.so.6
0000737b4cbc0000 312 120 0 r---- libc.so.6
0000737b4cc0e000 16 16 16 r---- libc.so.6
0000737b4cc12000 8 8 8 rw--- libc.so.6
0000737b4cc14000 52 20 20 rw--- [anon]
0000737b4cce4000 12 8 8 rw--- [anon]
0000737b4ccf7000 8 4 4 rw--- [anon]
0000737b4ccf9000 16 0 0 r---- [anon]
0000737b4ccfd000 8 0 0 r---- [anon]
0000737b4ccff000 8 8 0 r-x-- [anon]
0000737b4cd01000 4 4 0 r---- ld-linux-x86-64.so.2
0000737b4cd02000 188 180 0 r-x-- ld-linux-x86-64.so.2
0000737b4cd31000 44 44 0 r---- ld-linux-x86-64.so.2
0000737b4cd3c000 8 8 8 r---- ld-linux-x86-64.so.2
0000737b4cd3e000 4 4 4 rw--- ld-linux-x86-64.so.2
0000737b4cd3f000 4 4 4 rw--- [anon]
00007ffffe931000 132 12 12 rw--- [stack]
ffffffffff600000 4 0 0 --x-- [anon]
---------- ------- ------- -------
total kB 2772 1744 100
5901: ./cpu_bound


vboxuser@VM-1: ~ — top -p 5901
vboxuser@VM-1:~$ top -p 5901
top - 20:11:47 up 5 min, 1 user, load average: 2.19, 1.47, 0.72
Tasks: 1 total, 1 running, 0 sleeping, 0 stopped, 0 zombie
%Cpu(s): 11.7 us, 2.0 sy, 0.0 ni, 85.3 id, 0.0 wa, 0.0 hi, 0.9 si, 0.0 st
MiB Mem : 5406.8 total, 809.4 free, 1806.8 used, 2998.2 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 3599.9 avail Mem
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
5901 vboxuser 20 0 2636 1480 1388 R 100.0 0.0 4:00.05 cpu_bound